

Quantitative Empirical Methods Exam

Yale Department of Political Science, January 2025

You have 24 hours to complete the exam. The exam consists of three parts.

Back up your assertions with mathematics where appropriate and show your work. Good answers will provide a direct answer that illustrates an understanding of the question, and calculations or statistical arguments to validate the answer. Where applicable, exceptional answers will include all of these *as well as* proofs that are technically complete, including formally articulating sufficient assumptions and regularity conditions. Questions will not be weighted equally. A holistic score will be assigned to the exam. Therefore, it is important to demonstrate your understanding of the material to the best of your ability.

Part 1 (Theoretical section) consists of short questions that can be answered with pen and paper. You are allowed to consult textbooks and other reference material, but the questions are written so that well-prepared students should be able to answer them without such references.

Part 2 (Essay section) contains a recent, well-regarded empirical article. We will ask you to offer an evaluation of its methodological approach and presentation of results. In particular, we will advise you to pay particular attention to the identification conditions (either explicit or implicit), the associated estimation strategy, and possible threats to inference. Your response may be anywhere from 500 to 1,000 words.

Part 3 (Computer assisted section) will involve using statistical software to answer one longer exercise with several associated questions. A complete answer to Part 3 will include code and output, as well as your written answers.

For the whole exam, you are permitted access to any and all written materials, as well as unrestricted use of your own computer with access to the internet. The only restriction is that you may **not** interact with any person, online or otherwise.

Please turn in your answers as an email to colleen.amaro@yale.edu.

1 Theoretical section

1. Provide brief definitions of the following terms:
 - (a) bias
 - (b) confidence interval
 - (c) bootstrap
 - (d) p -value
 - (e) statistical power
 - (f) consistency
2. State if the following statements are true/false. In either case, briefly explain using an example or an equation why the statement is true/false.
 - (a) There is a 95% probability that the true population parameter is within an estimated 95% confidence interval.
 - (b) A p -value of 0.1 means that there is a 10% probability that the null hypothesis is true. 10% is not very low, so we fail to reject the null hypothesis, since it could be true.
 - (c) Because of Weak Law of Large Numbers, when you have a sufficiently large sample, you no longer need to worry about bias in estimating population quantities.
 - (d) When we reduce the size of a hypothesis test, we reduce the number of potential findings that we falsely conclude are statistically significant. This is done by reducing α to a lower value.
 - (e) Central Limit Theorem ensures that every random sample we draw will be Normally distributed.
 - (f) If the correlation between two random variables is 0, this implies these two variables are independent of one another.
3. Fifty draws are made at random with replacement from a box. Classify each of the following as: a parameter, an observed value, or nonsense. Why?
 - (a) The variance of the average of the 50 draws.
 - (b) The average of the variance of the 50 draws.
 - (c) The average of the 50 draws.
 - (d) The variance of the 50 draws.
4. What is the distribution of $\log(U) - \log(1 - U)$ where U is uniform on $[0,1]$?

5. You want to estimate the average impact of historic exposure to violence on support for a peace referendum that took place in 2016. You collect a dataset of individual-level demographic covariates, like partisanship, gender, family income, education, and the district an individual lives in. You also collect a dataset with district-level covariates, like GDP, elevation, urbanicity, and counts of violent incidents (2005 to the present, 2024). You define a treatment indicator T_i that takes on a value of 1 if a district has a historic count of violent incidents from 2011-2015 greater than 0 and a value of 0 if the district had no violent incidents, and the outcome Y_i as the proportion of individuals in district i who voted in favor of the peace referendum.
 - (a) What is your unit of analysis?
 - (b) Given how Y_i and T_i are defined, define what $Y_i(1)$ and $Y_i(0)$ would be in the potential outcomes framework.
 - (c) Your colleague suggests you run a regression, regressing the outcome Y_i on the treatment T_i , and interpreting the coefficient in front of the treatment as the causal effect of historic exposure to violence on support for peace. Explain why this will likely not provide a valid estimate of the causal effect of historic exposure to violence on support for peace.
 - (d) Instead of running a regression, you decide to perform a matching analysis, where you will match on the covariates you have collected. Which covariates do you include? Why? What assumptions must hold for your matching analysis to provide a valid estimate of the treatment effect of interest?
 - (e) You begin with exact matching. You find that about 60% of the observations in your dataset cannot be matched. You estimate a matched difference-in-means. What identification assumptions must hold for this to provide an unbiased estimate of your treatment effect of interest?
 - (f) You decide to use inexact matching instead. What are examples of inexact matching approaches? What assumption must hold for this approach to provide an unbiased estimate of your treatment effect of interest?
 - (g) Instead of using matching, explain an alternative estimation approach you could use to estimate the treatment effect of interest. What are the assumptions needed for the approach you propose?
6. We consider the roll-out of mandatory masking policies during COVID 19 for some congressional districts, but not others. For $t \geq 0$, $Z_{it} = 1$ for districts in the treatment group (a mandatory masking policy was adopted), and $Z_{it} = 0$ for the control districts. For $t < 0$, $Z_{it} = 0$ for everyone. Time t is measured in increments of weeks, and i corresponds to a specific district. Assume for simplicity that once a district institutes a masking policy, the masking policy remains in place. We're interested in the effect of masking policies on anti-government sentiment, as measured through

the number of anti-government Tweets sent from congressional district i , Y_{it} (for our purposes, assume that we can geocode every tweet).

- (a) If Z_{it} has been randomly assigned, how could we compute the impact of masking policies on anti-government sentiment?
- (b) We decide to use a difference-in-differences model. What parametric model would allow you to identify the effect of masking, given two time periods.
- (c) Write out the regression for how you would estimate a single treatment effect in the post-period for the treatment.
- (d) What are some checks you can conduct to evaluate the credibility of the underlying assumptions?
- (e) Describe in words how having a staggered roll-out provides a more robust identification approach.
- (f) What issues might arise if you ran the same regression as in part c) above with the staggered roll-outs? You can use math, but make sure to also explain the intuition behind your answer.

2 Essay section

Read the following article and offer a critical evaluation of its methodological approach and presentation of results. Note: “critical” does not imply that you must only criticize – it is recommended that you give credit to the authors if and when their arguments are convincing and/or novel with respect to standard practice.

Your response should be between 500 to **1000** words. We advise you to pay particular attention to the identification conditions (either explicit or implicit), the associated estimation strategy, possible threats to inference, statistical power, and generalizability. Justify each of your claims and, where applicable, suggest ways in which this line of research might be improved. We do not expect you to have special expertise in the topic area, but we do expect you to bring to bear your general analytical skills as a political scientist.

Article:

Grier, Kevin, Robin Grier, and Henry J. Moncrieff. “Uncertain times: The causal effects of coups on national income.” *American Journal of Political Science* (2024). <https://doi.org/10.1111/ajps.12884>

3 Computer assisted section

Begin by creating two random variables in R using the following code:

```
set.seed(02139)
X1 <- rnorm(100000, 5, 2)
X2 <- rexp(100000, 0.2)
```

For the purposes of this problem, you should treat these variables (each with 100,000 elements) as the full population. You will take samples from these two datasets to evaluate the coverage probability of 95% confidence intervals for the population mean using different types of data and different sample sizes.

1. Plot and describe the full distributions of X1 and X2. What is the population mean and population standard deviation of each random variable?
2. Now, create a loop to take 100 samples of size 8 from each dataset and record the sample mean for each sample. Plot the density of sample means for X1 and X2 separately and compare these densities. Are they similar or different? Why?
3. Given your answer in Question 1, what should be standard error of the sample mean of X1 and the standard error of the sample mean of X2?
4. We know that if a random variable is normally distributed, 95% of its distribution will be within 1.96 standard deviations of the mean. What proportion of the sample means for X1 and X1 are within 1.96 standard errors of the population means?
5. Repeat the simulations in Questions 2-4 for samples of sizes 8, 20, 50, and 500 using both 100 and 1,000 simulations. Report the probability of being within 1.96 standard errors of the population mean for each of your eight simulations in a table (you do not need to create additional plots). How do your results change? What differences do you see between X1 and X2?
6. How does the central limit theorem explain what you see in the simulations?